T.J.S. ENGINEERING COLLEGE

(AN AUTONOMOUS INSTITUTION)

Approved by AICTE, Affiliated to Anna University, Chennai.

TJS Nagar, Peruvoyal, Kavaraipettai, Chennai-601206.



M.E. VLSI DESIGN

REGULATIONS 2026

CURRICULUM

(For the Students admitted from 2026-27)

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REGULATIONS 2026

CHOICE BASED CREDIT SYSTEM (CBCS)

M.E. VLSI DESIGN

Course Type	Definition
T	Theory
LIT	Lab Integrated with Theory
L	Laboratory

Period	Definition
L	Lecture Period
T	Tutorial Period
P	Practical Period

Course Category	Definition
BS	Basic Science Course
PC	Professional Core Course
PE	Professional Elective Course
OE	Open Elective Course
SD	Skill Development Course

1 Hour Lecture (L) per week	1 Credit
1 Hour Tutorial (T) per week	1 Credit
1 Hour Practical (P) per week	0.5 Credit

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SEMESTER I

S.No	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26MA501	Graph Theory and Optimisation Techniques	T	3-1-0	4	4	BS
2	26VL502	Analog Integrated Circuit Design	T	3-0-0	3	3	PC
3	26VL601	Digital CMOS VLSI Design	LIT	2-0-2	4	3	PC
4	26VL602	Advanced Digital System Design	LIT	3-0-2	5	4	PC
5	26VL603	ASIC and FPGA Design	LIT	3-0-2	5	4	PC
6	26VL701	Analog Integrated Circuit Design Laboratory	L	0-0-4	4	2	PC
7	26SD801	Technical Seminar	-	--	--	1	SD
				TOTAL	**26**	**21**	

SEMESTER II

S.No	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26VL503	Multicore Architecture and ISA Verification	T	3-0-0	3	3	PC
2	26VL504	VLSI Testing and Verification using UVM	T	3-0-0	3	3	PC
3	26VL604	Data Converters IC Design	LIT	2-0-2	4	3	PC
4	26VL605	Low Power VLSI Design	LIT	2-0-2	4	3	PC
5		Professional Elective-I	T / LIT			3	PE
6	26VL702	Verification using UVM Laboratory	L	0-0-4	4	2	PC
7	26SD802	Industry Oriented Course-I	-	1-0-0	1	1	SD
8	26SD803	Industrial Training	-	-	-	1	SD
9	26SD804	Self Learning Course	-	-	-	1	SD
				TOTAL		**20**	

SEMESTER III

S.No	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1		Professional Elective-II	T/LIT			3	PE
2		Professional Elective-III	T/LIT			3	PE
3		Professional Elective-IV	T/LIT			3	PE
4		Professional Elective-V	T/LIT			3	PE
5	26SD805	Industry Oriented Course-II				1	SD
6	26PW801	Project Work-I	L	0-0-12	12	6	SD
				TOTAL		**19**	

SEMESTER IV

S.No	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26PW802	Project Work-II	L	0-0-24	24	12	SD
				TOTAL	**24**	**12**	

CREDIT DISTRIBUTION

Course Category	I	II	III	IV	TOTAL
BS	**4**				**4**
PC	**16**	**14**			**30**
PE		**3**	**12**		**15**
SD	**1**	**3**	**7**	**12**	**23**
	21	**20**	**19**	**12**	**72**

PROFESSIONAL ELECTIVE COURSES							
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26VL606	VLSI Signal Processing	LIT	2-0-2	4	3	PE
2	26VL607	RF Integrated Circuit Design	LIT	3-0-0	3	3	PE
3	26VL505	Electromagnetic Interference and Compatibility	T	3-0-0	3	3	PE
4	26VL506	Physical Design of VLSI	T	3-0-0	3	3	PE
5	26VL507	Hardware / Software Co-Design	T	3-0-0	3	3	PE
6	26VL608	CAD for VLSI Design	LIT	3-0-0	3	3	PE
7	26VL508	Nanotechnology	T	3-0-0	3	3	PE
8	26VL509	Pattern Recognition	T	3-0-0	3	3	PE
9	26VL510	Signal Integrity for High Speed Design	T	3-0-0	3	3	PE
10	26VL511	Network on Chip	T	3-0-0	3	3	PE
11	26VL512	VLSI for Wireless Communication	T	3-0-0	3	3	PE
12	26VL513	DSP Structures for VLSI	T	3-0-0	3	3	PE
13	26VL514	MEMS and NEMS	T	3-0-0	3	3	PE
14	26VL515	Advanced Transistor Modeling	T	3-0-0	3	3	PE
15	26VL609	VLSI Architecture for Image Processing	T	3-0-0	3	3	PE
16	26VL516	Reconfigurable Architecture	T	3-0-0	3	3	PE
17	26VL517	Adaptive Signal Processing	T	3-0-0	3	3	PE
18	26VL518	Digital Image and Video Processing	T	3-0-0	3	3	PE
19	26VL519	Evolvable Hardware	T	3-0-0	3	3	PE